Appendix to How Does State Violence Experienced During Adolescence Shape State–Person Narratives?

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Guide to the Appendix

This appendix follows the same order in which the main text sends the reader outward. The measurement appendix gathers the coding, threshold, scoring, aggregation, and covariate details promised in the data and measurement sections. The robustness appendix then gathers the supplementary checks promised in the model, results, and robustness sections. Table 1 is the quickest map from a main-text appendix reference to the relevant appendix location, and it serves as the only appendix-wide routing table. The opening subsections of Appendix A and Appendix B are therefore brief local guides rather than second full roadmaps.

Table 1: Reader's guide to the appendix.

Main-text section	Reader question	Appendix location
The Densho Digital Repository and the Study Population	How are the grouped treatment bins, recovered baseline covariates, and descriptive coding choices constructed?	Sections A.2 and A.6
Treatment, Outcome, and Covariates	What exact scoring settings, threshold values, boundary rules, multimodal fusion rules, ILR mirrors, and aggregation choices sit behind the main-text outcomes?	Sections A.3, A.4, and A.5
Treatment, Outcome, and Covariates; Robustness Checks and Scope Limits	Which appendix checks revisit threshold choice, aggregation, ILR representation, observability, and recovered-unknown handling?	Sections B.3 and B.5
Causal Estimation via Double/Debiased Machine Learning	Which checks bear on overlap, omitted confounding, transport scope, placebo falsification, and single-site leverage?	Sections B.2 and B.4
Treatment Effects on Narrative Stance and Composure	Where is the appendix-side companion to the main results tables, including ILR mirrors and the archived companion result inventories?	Sections B.3 and B.8
Robustness Checks and Scope Limits	Where do the main-text alternative-coding, restricted-severity-bundle, and complete-case discussions expand into the fuller appendix battery?	Sections B.2, B.4, and B.5

A Measurement Appendix

This appendix records implementation details deferred from the main text. It is organized in the same order in which the body of the paper points readers here: first the treatment and covariate coding choices referenced in the data and measurement sections, then the scoring, threshold, and aggregation rules behind the main-text outcomes, and finally the analysis-frame conventions that matter for robustness and scope. Where the main text points to an appendix-side measurement or threshold check, the relevant robustness subsection is cross-referenced explicitly.

A.1 How to Use the Measurement Appendix

Table 1 is the appendix-wide routing table. Appendix A has a narrower purpose: it records how the paper's treatments, outcomes, and recovered baseline covariates are constructed. Readers who want grouped timing, dosage, severity, and recovered-baseline coding should start with Sections A.2 and A.6. Readers who want exact scoring rules, threshold values, and narrator-level outcome assembly should start with Sections A.3–A.5. When the question shifts from how a measure is defined to whether that definition is robust, the relevant follow-on sections are Section B.3 for thresholds, aggregation, and ILR representation and Section B.5 for observability and archiveability concerns.

A.2 Treatment Construction and Descriptive Groupings

This subsection is the measurement-appendix counterpart to the coding notes in the descriptive table and to the treatment definitions in the main measurement section.

Treatment coding details. Timing is constructed from narrator birth year and first recovered facility assignment. The main grouped categories reported in the main text are childhood exposure, adolescent exposure, and adult exposure. The main binary timing

specification uses the broader adolescence band of ages 10 to 18. The alternative timing check reported in the main text instead partitions first exposure into childhood (before age 13), adolescence (ages 13 to 17), and adulthood (age 18 or older). Alternative narrow bands retained for sensitivity checks are early adolescence (10 to 13), middle adolescence (14 to 16), and late adolescence (17 to 18).

Dosage is measured as total days incarcerated across all recovered confinement spells. In the main descriptive table, dosage is shown in empirical quartiles (Q1–Q4) for compact descriptive presentation. The analysis itself retains the continuous incarceration-days measure, so the appendix does not repeat the grouped counts here.

Severity is based on four binary indicators: family separation, parental arrest, loyalty-questionnaire conflict, and segregation-camp assignment. The appendix distinguishes between the descriptive count used for tabulation and the normalized aggregate score used in the main causal tables. For descriptive presentation, the paper reports the additive count,

$$\text{severity_index_v1} = \text{family separation} + \text{parental arrest} + \text{loyalty conflict} + \text{segregation}.$$

The grouped severity categories reported in the main text are:

Category	Definition
No severe betrayal markers	severity = 0
Low severity	severity = 1
Moderate severity	severity = 2
High severity	severity ≥ 3

The main treatment results in the paper, however, use the normalized aggregate severity score `severity_index_v2`. Its component weights are documented here so that they need not appear in the main text:

$$\text{severity_index_v2} = \frac{\begin{array}{c}1.5(\text{family separation}) + 1.5(\text{parental arrest}) \\ + 1.0(\text{loyalty conflict}) + 2.0(\text{segregation})\end{array}}{\text{normalizing constant}}.$$

This normalized score is the aggregate severity measure that appears in the main causal tables. The additive count remains in the paper because its grouped values are easier to interpret descriptively, while the normalized score is the version carried into the joint causal models.

A.3 Scoring Procedure and Canonical Analytic Base

This subsection corresponds to the places in the main measurement section that defer exact scoring settings, the canonical analytic base, and related provenance notes to the appendix.

Segment scoring details. Each transcript segment is scored by GPT-5.4 using the full segment text as input. The final paper uses deterministic settings. The text scorer emits an authority-stance score, a belonging-stance score, and a text-side composure score. The paper's final how specification then fuses that text-side score with the archived audio/video run already stored under `data/processed/how_av_scores_v1/`.

Final paper implementation parameters:

Parameter	Value
Model	GPT-5.4
Temperature	0.0
Input unit	Full transcript segment text
Status	Multimodal final-paper outcome scoring specification

Canonical analytic base. The final paper's v2 reclassification is built from the canonical generic segment source rather than from the earlier broader model-specific files. Con-

cretely, the analytic segment input is `segment_llm_fulltext_2d_outcomes_v1.csv`, with interview- and narrator-level companion files `interview_llm_fulltext_2d_outcomes_v1.csv` and `narrator_llm_fulltext_2d_outcomes_v1.csv`. The canonical segment file is the intersection-compatible segment corpus used to construct the generic interview and narrator tables. After the 2026-04-17 realignment, this analytic base contains 14,390 segments, 729 interviews, and 642 narrators. Legacy model-specific outputs, chi-1 rerun artifacts, and superseded wrong-base v2 files are retained only under the dated archive root documented in the appendix lineage notes.

A.4 Outcome Thresholds, Boundary Rules, and Representation

This subsection records the boundary map and multimodal threshold rules that the main text refers to but does not print in detail. The appendix-side robustness counterparts are the threshold and representation checks in Section B.3.

What-dimension boundary map. The GPT scorer used in the paper places narrator-level scores into a two-dimensional authority-stance by belonging-stance space. The final paper boundary values are:

Boundary	Value
Authority low cut	0.50
Authority high cut	0.54
Belonging cut	0.43

These boundaries define six regions that collapse to three narrator-level what categories:

Three-class outcome	Merged regions
Injury	injury + injury_disrupted
Rupture	adaptive_rupture + default_rupture
Distrust	politicizing_rupture + distrust

The three-class distribution reported in the main descriptive table. The appendix records only the boundary map and merge rule used to produce it.

Boundary sensitivity checks indicate that the final paper's three-class distribution is not driven by a single arbitrary threshold choice. The full grid and decision memo remain archived in the project materials. The direct appendix-side test of this claim is the boundary-sensitivity check in Section B.3, especially R6 in Table 3.

How-dimension multimodal rule. In the final paper specification, the retained variable name `composure_score` refers to the text-side discomposure subscore, with 0 denoting fully composed narration and 1 denoting fully discomposed narration. Let $c_s^{text} = \texttt{composure_score}_s$. The audio and video archives provide calibrated composite z-scores z_s^{audio} and z_s^{video}. The paper's main specification keeps the text threshold at 0.50 and refreezes the archived audio/video z-composite distributions at their full-corpus marginal 80th percentiles, 0.587275 and 0.372174. We center those modality-specific thresholds and map them to $[0, 1]$ using the standard normal cumulative distribution function:

$$c_s^{audio} = \Phi\big(z_s^{audio} - 0.587275\big), \qquad c_s^{video} = \Phi\big(z_s^{video} - 0.372174\big).$$

The multimodal how score used in the paper is the strongest available modality signal,

$$c_s^{mm} = \max\{c_s^{text}, c_s^{audio}, c_s^{video}\},$$

and the binary split used in the paper is:

$$\text{discomposed} = 1 \quad \text{if } c_s^{mm} \geq 0.50.$$

Because the six-cell mean shares belong in the main descriptive table, the appendix records only the classification rule and the aggregation logic below. Under this specification, the binary label is equivalent to an OR-gate across the text, audio, and video

modality indicators because each modality-specific score crosses 0.5 exactly at its assigned threshold. On the full archived A/V corpus, this main-analysis rule implies marginal discomposure rates of 20.0 percent for audio and 19.5 percent for video, with an A/V OR-rate of 34.72 percent before the text arm is folded back in.

As a robustness check, the project also preserves the text cutoff at 0.50 while retuning only the audio and video thresholds to recover the archived A/V union target of 0.323. The resulting robustness pair is 0.490034 for audio and 0.533734 for video, which yields full-corpus A/V rates of 24.00 percent and 12.19 percent and an OR-rate of 32.28 percent. The main paper uses the marginal-80th-percentile refreeze as the primary specification and treats the union-target pair as a threshold-governance robustness analysis. That threshold-governance question is taken up again in the measurement-robustness discussion in Section B.3.

A.5 Narrator-Level Outcome Assembly

This subsection is the appendix-side companion to the main text's discussion of narrator-level shares, ILR mirrors, and alternative aggregation choices.

Narrator-level aggregation details. Let $(s_1, \ldots, s_6)$ denote the narrator's share vector over the six cells of the what-by-how typology, ordered as Guarded Vigilance, Diffuse Unease, Wounded Belonging, Stigmatized Discomposure, Oppositional Citizenship, and Ambivalent Estrangement, with

$$\sum_{k=1}^{6} s_k = 1.$$

These shares form the compositional outcome used in the first estimation track. In the main text, the six named what-by-how shares are the reported main-text outcomes. For appendix-side technical mirrors, the same ordered six-part composition is also mapped into five isometric log-ratio coordinates using the standard pivot representation,

$$z_j = \sqrt{\frac{j}{j+1}} \ln\left(\frac{\left(\prod_{k=1}^{j} s_k\right)^{1/j}}{s_{j+1}}\right), \qquad j = 1, \ldots, 5.$$

This transformation respects the simplex constraint while placing the same compositional outcome in $\mathbb{R}^5$ for appendix-side technical mirrors and robustness displays.

The second estimation track uses narrator-level averages of three continuous scores,

$$\bar{a}_i, \quad \bar{b}_i, \quad \bar{c}_i,$$

where $\bar{a}_i$ is mean authority-stance for narrator i, $\bar{b}_i$ is mean belonging-stance, and $\bar{c}_i$ is mean multimodal how score.

Alternative aggregation checks. The main text aggregates all eligible scored segments across all eligible interviews for each narrator. Because some narrators appear in multiple interviews, robustness checks compare this all-interviews specification against simpler alternatives such as earliest interview only, primary interview only, and one-interview-per-narrator constructions. The appendix-side reader guide to those checks is Section B.3: R7 revisits aggregation, R9 revisits the ILR mirror, and the comparison of the two outcome tracks returns to whether those two summaries carry the same substantive meaning.

A.6 Covariate Conventions and Analysis Frames

This final subsection gathers the covariate conventions and analysis frame choices that the main text sends to the appendix rather than to the body tables.

Covariate normalization and missing-data conventions. Generation is normalized to the four-category set Issei, Nisei, Kibei, and Sansei. Prewar place is grouped to West Coast, inland, and Hawaii for the main text. Interview year is drawn from the archive metadata, and age at interview is defined as interview year minus birth year. The re-

covered pre-treatment baseline block is limited to family SES, prewar education, prewar English ability, and cultural orientation, together with supporting factual indicators for Japan-linked background. Only clearly prewar evidence is admissible for those recovered variables; wartime experience, postwar adjustment, and later-life political reflection are excluded from baseline coding. Rows with structural treatment non-observability remain outside the treatment-observed analytic sample. Remaining item-level missingness in the analytic sample is low in the core archive block and is reported descriptively in the main text.

Recovered pre-treatment baseline procedure. Recovery proceeds in two passes. We first build a deterministic narrator-level baseline from structured sources already linked to the archive record, including the WRA registry extract, narrator-page metadata, and prewar segment-summary evidence. This pass can assign supported values for family SES, prewar education, prewar English ability, cultural orientation, and the auxiliary Japan-linkage flags, but unsupported fields remain explicit `unknown`. We then build one narrator packet per treatment-observed case and use transcript- backed prewar evidence from interview summaries and full-transcript retrieval to recover remaining gaps. Only admissible prewar evidence is used in this second pass; wartime experience, postwar adjustment, and later-life political reflection are excluded unless they directly quote prewar conditions. In the final merge, non-`unknown` transcript-based returns override the deterministic baseline, the first pass is used only when the second pass remains `unknown` or is unavailable, and disagreements are retained with conflict flags and a manual-review recommendation so that the final frame remains auditable. The detailed extraction protocol and runbook are archived in `appendix/pretreatment_covariate_extraction_contract_v1.md` and `appendix/pretreatment_covariate_extraction_runbook_v1.md`.

Recovered pre-treatment baseline distribution. The final merged pretreatment frame is the project file `pretreatment_covariate_final_frame_v1.csv`. Table summaries are

reported here for the two cohorts that matter for downstream modeling: `treatment_observed` ($N = 688$) and the outcome-defined main modeling sample ($N = 633$).

Variable	Treatment-observed ($N = 688$)	Main modeling ($N = 633$)
Family SES	small business 38.5%; farming/fishing 37.5%; professional white collar 8.7%; wage/service 7.6%; unknown 7.7%	small business 38.4%; farming/fishing 37.9%; professional white collar 9.3%; wage/service 7.9%; unknown 6.5%
Education	secondary 45.9%; primary or less 25.6%; schooling unspecified 13.4%; college plus 10.0%; unknown 5.1%	secondary 47.4%; primary or less 25.6%; schooling unspecified 13.3%; college plus 10.1%; unknown 3.6%
English ability	bilingual/functional 68.2%; child not applicable 23.3%; limited English 4.2%; Japanese only/none 0.9%; unknown 3.5%	bilingual/functional 68.2%; child not applicable 23.2%; limited English 4.6%; Japanese only/none 0.8%; unknown 3.2%
Cultural orientation	mixed Japan-US 74.0%; US embedded 23.7%; Japan embedded 2.2%; unknown 0.1%	mixed Japan-US 76.5%; US embedded 21.5%; Japan embedded 2.1%; unknown 0.0%

Statistic	Treatment-observed	Main modeling
Any recovered `unknown` across the four categorical fields	90 (13.1%)	69 (10.9%)
Recovered complete-case rows	598 (86.9%)	564 (89.1%)
Born in Japan = 1	18 (2.6%)	16 (2.5%)
Time in Japan = 1	347 (50.4%)	326 (51.5%)
Kibei = 1	31 (4.5%)	31 (4.9%)
Japanese language school = 1	254 (36.9%)	245 (38.7%)

Recovered `unknown` handling. The main causal specification keeps recovered `unknown` values as explicit categories for the four recovered categorical baseline fields: family SES, prewar education, prewar English ability, and prewar cultural orientation. This preserves the full outcome-defined main estimation sample at $N = 633$. A separate recovered complete-case rerun is reserved for robustness and drops narrators with any remaining `unknown` in those four fields, reducing the estimation sample to $N = 564$. The upstream treatment/covariate frame contains 670 rows, but 37 of those rows lack canonical v2 outcomes and therefore do not enter the final estimand. No multiple imputation is used for the 18 treatment-observed rows outside that upstream frame, and no multiple imputation is used for recovered `unknown` states inside the main estimation sample. The causal consequence of this handling rule is reported in Section B.5, especially the recovered complete-case rerun in Table 5.

B Robustness Appendix

This appendix reports the broader robustness checks for the baseline narrator-level specifications. Two patterns survive the largest share of the battery. First, the severity index remains positively associated with authority stance and with the joint distrust-discomposure outcome. Second, the broad binary timing indicator remains weaker and more specification-sensitive. For readability, the discussion below uses the paper's own labels rather than internal variable names. The appendix is organized by reader question rather than by the order in which the checks were run so that each place where the main text points outward lands on a visible appendix counterpart.

B.1 How to Use the Robustness Appendix

Table 1 remains the appendix-wide route map. Appendix B itself is organized by type of robustness question. Section B.2 covers timing recoding, reweighting, and support. Sec-

tion B.3 covers threshold, aggregation, minority-signal, and ILR representation checks. Section B.4 covers omitted-confounding diagnostics, the restricted severity bundle, placebo permutations, and site leverage. Section B.5 covers recovered complete-case handling, negative controls, observability, and scope. Section B.6 gathers extensions. Section B.8 then expands R1–R19 into one review table per numbered check.

B.2 Timing, Weighting, and Support Checks

This first block corresponds to three different appendix promises in the main text. It is where readers should look if they want to know whether the weak adolescent-dummy result is merely a cutpoint artifact, whether the weighted results depend on one transport convention, or whether the joint model relies on weak overlap in the treatment-support space. R1 and R2 ask whether one nuisance-learner choice or one fold partition drives the result. R3 revisits treatment coding, including the alternative timing-and-dosage coding discussed in the main text. R4 addresses overlap and positivity through trimming. R5 addresses the dependence of the main claims on a particular weighting convention.

The alternative timing check is the part of this section most directly connected to the main text. Adult-versus-adolescence contrasts remain weak, whereas childhood-versus-adolescence contrasts are larger on broader rupture and composed-narrative summaries. That is why the main robustness section treats timing as a boundary question rather than as a new headline effect. Learner sensitivity and cross-fitting then show that the main severity result is not being manufactured by one nuisance-model configuration or one fold partition.

The remaining checks are more substantively informative. Disaggregating the severity bundle shows that no single component reproduces the full composite pattern by itself, although loyalty conflict is a recurring contributor on both authority stance and the joint distrust-discomposure outcome. Overlap trimming changes some magnitudes, but the main severity rows remain positive across weighting schemes. Likewise, moving be-

tween inverse-probability weighting, raking, and unweighted estimation does not alter the central authority-side and distrust-side severity pattern, even though secondary belonging and rupture outcomes are less uniform.

Table 2: Timing, weighting, and support checks.

Check	What it tests / alternative specification	Headline evidence	Interpretation
R1	Learner sensitivity	Same-sign rate 83.7%; median absolute drift 2.339×10^{-5}; severity on authority stance moves from 0.0409 in the baseline to 0.0547 under lasso and 0.0499 under random forest; severity on joint distrust-discomposure moves from 0.1393 to 0.1549 and 0.1316	The main severity pattern is not driven by one learner choice; timing is less stable
R2	Cross-fit folds	Same-sign rate 90.1%; median absolute drift 8.143×10^{-6}; severity rows remain within very narrow ranges across 3-, 5-, and 10-fold estimation	The main severity pattern is not driven by one fold partition
R3	Three-category timing, dosage bins, and severity components	Adult-minus-adolescence contrasts stay weak, whereas childhood-minus-adolescence rises to 0.104, 0.141, and 0.158 on the broader-rupture summary; loyalty conflict remains the clearest severity component on authority stance and on the joint distrust-discomposure outcome	The weak binary timing result is not a simple cut-point artifact; timing is more developmentally differentiated, and severity remains concentrated in selected components
R4	Overlap trimming	Sample falls to 561–562 observations; same-sign rate 76.5%; median absolute drift 0.008948; the main severity rows remain positive across weights	Support restriction changes magnitudes, but not the core authority/distrust conclusion
R5	Weight specification	Across IPW, rake, and unweighted estimation, severity remains positive on authority stance and on the joint distrust-discomposure outcome	The main severity result is weighting-robust; secondary outcomes are less uniform

B.3 Measurement, Threshold, Aggregation, and Representation Checks

This block is the appendix counterpart to the repeated appendix notes in the measurement section. R6 asks whether the what-side boundary map is arbitrary. R7 asks whether the preferred all-interviews narrator aggregation rule matters. R8 asks whether the headline result depends on low-frequency outcome cells. R9 asks whether the main-text PLR representation is contradicted by the ILR geometry-respecting mirror. The comparison

of the two outcome tracks then asks whether the two outcome summaries move in the same substantive direction. The general pattern is that the measurement system is not arbitrary, but some margins are more stable than others.

The boundary-sensitivity exercise is the clearest example. Around the baseline cutpoint, small changes in one boundary parameter leave the class shares unchanged, while moving the high-side cutpoint shifts observations between rupture and distrust. This suggests that the baseline classification is not perched on an unstable knife-edge, but it also shows why the fine rupture-versus-distrust split should remain an appendix-level result rather than a heavily emphasized headline claim.

The remaining checks reinforce this pattern. Alternative aggregation is a hard stress test because it sharply reduces the effective sample and changes the analysis unit; its mixed performance therefore speaks more to estimand and sample changes than to a routine robustness failure. By contrast, the minority-signal and ILR dimensionality checks do not overturn the main interpretation. The comparison of the two outcome tracks is strongest for authority/distrust and discomposure-linked constructs, and noticeably weaker for belonging-side contrasts. That is the right way to summarize measurement robustness in this setting: strong on the principal authority and distrust dimensions, weaker on finer compositional and belonging-side margins.

B.4 Identification-Assumption, Placebo, and Leverage Checks

This block corresponds most directly to the identifying assumptions in the model section. Conditional unconfoundedness is probed most directly by R10 and R11. Overlap and positivity are addressed by R4 in the preceding subsection. The transport-scope question is partly probed by R5 above. R12 asks whether the narrator-level estimand looks different under an interview-level reformulation. R13 asks whether the pattern of estimates depends on simultaneous rather than sequential orthogonalization. R14 is a falsification check using placebo assignments. R15 asks whether one camp is driving the headline re-

Table 3: Measurement, threshold, aggregation, and representation checks.

Check	What it tests / alternative specification	Headline evidence	Interpretation
R6	Boundary sensitivity	Baseline upper and lower cutpoints are 0.54 and 0.43; baseline shares are 36.8% Injury, 50.0% Rupture, and 13.2% Distrust; lowering the upper cutpoint to 0.53 moves 7 percentage points from Rupture to Distrust	The baseline boundary is locally stable, but fine rupture-versus-distrust allocation is threshold-sensitive
R7	Alternative aggregation	Effective analysis falls to 31 narrators; same-sign rate 54.0%; median absolute drift 0.01106	This is a demanding stress test, but too disruptive to replace the preferred narrator-level design
R8	Minority-signal outcomes	Same-sign rate 100%; zero coefficient drift on the highlighted baseline rows	The main severity pattern does not depend on dropping low-frequency outcome cells
R9	ILR dimensionality	63 rows; 7 raw $p < 0.05$ rows; no headline contradiction identified in the review	The dimensionality choice does not force a revised interpretation
Cross-track comparison	Meaning consistency across the two outcome tracks	216 rows total; 130 same-sign and 86 opposite-sign rows overall; strongest agreement for authority/distrust and discomposure constructs	Cross-track support is strongest for authority/distrust and discomposure, and weaker for belonging-side contrasts

sult. These checks do not overturn the main conclusion, but they do narrow the language that can be used to describe it.

The omitted-confounding analysis is the most important caution. The main severity rows are not trivial, but they are not immune to omitted-variable concerns. The appropriate conclusion is not that the design is invulnerable, but that the strongest severity rows have non-negligible robustness values and remain economically meaningful under several other stress tests.

The rest of the battery is more reassuring. The restricted-severity exercise shows that the distrust-side pattern does not disappear when the severity bundle is decomposed, although no single component reproduces the full composite result. Sequential DML leaves the pattern of estimates almost unchanged. Leave-one-camp-out changes some magnitudes, but does not reduce the main authority-side or distrust-side severity result to a single-site artifact. By contrast, the interview-level long-format specification is too small and too unstable to serve as a preferred alternative, and the placebo exercise is supportive but mixed rather than uniformly dispositive.

B.5 Selection, Observability, and Scope

Selection checks matter most for two reasons. First, the baseline results should not be an artifact of recovered missingness handling. Second, the how dimension is more vulnerable than the what outcomes to archiveability and narratability concerns. The evidence in this section supports the first point clearly and the second point only partially. R16 is the direct check on recovered-unknown handling. The negative-control checks on the how dimension ask whether the how measure is being mechanically induced by noise or interview structure. The companion observability and MNAR checks show how much selection-on- observables and simple MNAR stressors can move the how effect, and the difficult-cases panel records boundary cases in a fixed appendix table rather than postponing them for a later hand-coded verdict.

Table 4: Identification-assumption, placebo, and leverage checks.

Check	What it tests / alternative specification	Headline evidence	Interpretation
R10	Omitted-confounding sensitivity	Median $RV_{q=1}$ across the panel is 3.2%; the authority-stance severity row has $RV_{q=1} = 0.1199$; the joint distrust-discomposure severity row has $RV_{q=1} = 0.1176$	The main severity rows are non-trivial, but not invulnerable to omitted confounding
R11	Restricted severity bundle	Total days incarcerated remains positive on authority stance with $p \approx 0.007$ to 0.008 across weights; family separation remains positive on the joint distrust-discomposure outcome and reaches significance in the unweighted fit	The main severity index is not reducible to one component, but the distrust-side pattern is not purely a compositional artifact
R12	Interview-level long format	37 interview observations representing 31 narrators; same-sign rate 52.4%	The interview-level reformulation is too unstable and low-power to replace the narrator-level design
R13	Sequential DML	27 rows; same-sign rate 96.3%; median absolute drift about 0.00075	The main pattern of estimates is essentially unchanged under sequential orthogonalization
R14	Placebo permutations	45 rows; 21 rows with empirical $p < 0.05$	The placebo exercise is supportive but mixed, not a universal pass/fail test
R15	Leave-one-camp-out	243 rows; same-sign rate 74.9%; median absolute drift 0.006656	Magnitudes are leverage-sensitive, but the main severity result is not a single-site artifact

The complete-case check preserves the main severity pattern. In the reduced sample, severity remains positive on authority stance and on the joint distrust-discomposure outcome across all three weighting schemes, while timing remains weak. This is strong evidence that the main severity result does not depend on recovered-unknown handling.

The how diagnostics are more informative than the earlier appendix note, but they remain mixed rather than exculpatory. In the negative-control check, adolescent exposure predicts a lower gap rate ($p = 0.015$) and dosage predicts shorter mean segment length ($p = 0.0037$), so some treatment variation co-moves with recording-related proxies. The trauma-versus-casual segment contrast is also not in the direction of a trauma-only discomposure story: trauma segments score about 0.008 lower on the multimodal how measure ($p = 0.0147$). At the same time, the base adolescent-exposure-to-how coefficient is small, and the interview-id residualization check strengthens rather than erases the negative association, which argues against a simple interviewer-driven artifact account.

The observability and MNAR checks tighten but do not eliminate the scope caution. Under the current observability rule, 430 of 642 narrators are classified as observable; among observed cases the mean predicted observability is 0.712, and the stabilized IPO weights top out at 1.548 after trimming. In the delta-adjusted MNAR exercise, shifting unobserved narrators by as much as 1.5 standard deviations toward greater discomposure moves the simple how ATE only from -0.0201 to -0.0203, while the Lee bound remains about -0.030. A companion difficult-cases panel also remains in the appendix: 69 narrators enter the panel, with 25 high-discomposure, 25 boundary-uncertain, and 19 technically poor but flagged cases. The appropriate scope condition is therefore narrower rather than deferred: the how findings remain interpretable for the observed archiveable narrator population, but they should not be overstated as if the negative-control and observability layers had fully closed narratability concerns.

Table 5: Selection and observability checks.

Check	What it tests / alternative specification	Headline evidence	Interpretation
R16	Recovered complete-case rerun	$N = 564$; severity on authority stance equals 0.049, 0.048, and 0.042 across IPW, rake, and unweighted estimation, all with $p < 0.001$; severity on the joint distrust-discomposure outcome equals 0.103, 0.106, and 0.090	The main severity result is not driven by recovered-missingness handling
Negative controls	How-side negative controls (treatment-to-noise, non-trauma contrast, and interview-id absorption)	Adolescent exposure predicts a lower gap rate ($p = 0.015$); dosage predicts shorter mean segment length ($p = 0.0037$); trauma-minus-casual multimodal how equals about -0.008 ($p = 0.0147$); interview-id residualization leaves the coefficient negative at -0.018 ($p = 0.0024$)	Negative controls do not support a pure interviewer-artifact account, but they are not clean nulls
Difficult cases	Difficult-cases panel	69 narrators included: 25 high-discomposure, 25 boundary-uncertain, and 19 technically poor but flagged, each with a segment package	The difficult-cases panel records boundary and low-quality cases without reducing them to a single hand-coded verdict
Scope	Observed archiveable sample plus observability/MNAR sensitivity	430 of 642 narrators satisfy the current observability rule; mean observed $\hat{p}(R_{how} = 1)$ is 0.712; stabilized IPO weights trim at 1.548; the delta-adjusted how ATE stays near -0.020 through $\delta = 1.5$; the Lee bound is about -0.030	Observability and MNAR adjustments do not overturn the sign of the simple how effect, but claims should remain scoped to the observed archiveable narrator population

B.6 Extensions and Heterogeneity

This appendix does not report a separate demographic subgroup battery. Instead, this subsection gathers supplementary developmental-amplification checks, the older embedding arm, interview-year interaction checks, and the composure-only specification. None of these checks displaces the baseline interpretation.

Supplementary developmental-amplification checks yield limited and non-uniform evidence. In Track 2, the adolescent-by-dosage interaction is negative on authority stance across the unweighted, inverse-probability, and raked estimands, but the adolescent-by-severity interaction remains weak across the displayed continuous outcomes. In Track 1, the interaction evidence is less stable: the Injury-share adolescent-by-dosage row is significant only in the unweighted fit and does not replicate under the transport estimands. Developmental amplification is therefore reported as a supplementary extension rather than as a coequal alternative to the main severity story.

The embedding arm is explicitly exploratory. It is based on a much smaller sample, it does not align closely with the baseline coefficients, and it should not be treated as a competing measurement system. Interview-year moderation is also weak: it does not convert the timing result into a robust moderated pattern. By contrast, isolating composure leaves the main how interpretation unchanged, which is reassuring because it shows that the how pattern is not mechanically generated by bundling composure together with the broader what-by-how structure.

B.7 Overall Interpretation

Taken together, these appendix checks support a bounded conclusion. Severity is the most stable treatment dimension in the current design, especially for authority stance and for the joint distrust-discomposure outcome. Timing remains weaker and more specification-sensitive. Belonging-side interpretation is more fragile than authority-side interpretation,

Table 6: Extensions and related checks.

Check	What it tests / alternative specification	Headline evidence	Interpretation
Amp.	Developmental-amplification interactions	Track 2 authority-stance adolescent×dosage = -1.42×10^{-5} ($p = 0.0257$), -1.70×10^{-5} ($p = 0.0174$), and -1.68×10^{-5} ($p = 0.0119$) across unweighted, IPW, and rake; adolescent×severity rows remain weak; Track 1 `share_class_Injury` adolescent×dosage is significant only unweighted	Amplification appears most clearly on authority stance, but does not generalize into a uniformly repeated cross-track pattern
R17	Embedding arm versus baseline language-model arm	31 observations; same-sign rate 46.7%; median absolute drift about 0.02048	The embedding arm is exploratory and too unstable to adjudicate the main results
R18	Interview-year interaction	Headline adolescent-by-interview-year rows remain weak or non-significant across the displayed baseline outcomes	Expressive-environment moderation does not rescue the weak timing result
R19	Composure-separate model	72 rows; same-sign rate 100%; zero median drift on the highlighted how rows	The how interpretation survives a cleaner composure-only specification

particularly once the analysis turns to alternative aggregation and cross-track comparison. The how dimension remains directionally supportive, but it continues to require narrower scope language because selection and narratability concerns are not fully resolved by the currently available diagnostics.

That is the most defensible summary of the appendix evidence. The battery does not overturn the paper's central result, but it does discipline how strongly each part of the argument can be stated.

B.8 Check-by-Check Direct-Evidence Inventory

Because the online appendix is not page-constrained, this final section expands the end inventory into one table per numbered robustness check. Each table reports the underlying result file, the focal within-check evidence, and the reading carried into the manuscript. The supplementary amplification interaction remains summarized in Table 6 rather than

repeated as a twentieth inventory table.

Table 7: R1. Learner sensitivity.

Full result table	Key rows and within-check statistics	Reading
`r1_learner_sensitivity_v1.csv`	Same-sign rate 83.7%; median absolute drift 2.339×10^{-5}; focal row: IPW authority-stance severity = 0.0409 in the baseline, 0.0547 under lasso, and 0.0499 under random forest	Learner choice does not overturn the main severity pattern

Table 8: R2. Cross-fit folds.

Full result table	Key rows and within-check statistics	Reading
`r2_crossfit_folds_v1.csv`	Same-sign rate 90.1%; median absolute drift 8.143×10^{-6}; focal row: IPW authority-stance severity = 0.0409 in the 5-fold baseline, 0.0387 in the 3-fold fit, and 0.0411 in the 10-fold fit	Cross-fitting partitions have negligible effects on the principal severity rows

Table 9: R3. Three-category timing, dosage bins, and severity components.

Full result table	Key rows and within-check statistics	Reading
`r3_treatment_discretization_v1.csv`	Focal contrasts: childhood-minus-adolescence rises to about 0.10, 0.14, and 0.16 on the broader-rupture summary, while adult-minus- adolescence remains weak; loyalty conflict on authority stance equals 0.0155 ($p = 0.016$), 0.0162 ($p = 0.012$), and 0.0185 ($p < 0.001$) across IPW, rake, and unweighted fits	The timing rerun is more developmental than the binary headline, and loyalty conflict remains a recurrent contributor without replacing the composite severity index

Table 10: R4. Overlap trimming.

Full result table	Key rows and within-check statistics	Reading
`r4_overlap_trimming_v1.csv`	Same-sign rate 76.5%; median absolute drift 0.008948; trimmed sample falls to 561–562 observations; focal result: the highlighted severity rows remain positive after overlap trimming	Support restriction moves magnitudes, but does not reverse the core authority/distrust result

Table 11: R5. Weight specification.

Full result table	Key rows and within-check statistics	Reading
`r5_weight_specification_v1.csv`	Focal rows: authority-stance severity = 0.0409 under IPW, 0.0381 under rake, and 0.0215 unweighted; joint distrust-discomposure severity = 0.1393, 0.1287, and 0.0788	The authority-side and distrust-side severity pattern survives reweighting; belonging remains less robust

Table 12: R6. Boundary sensitivity.

Full result table	Key rows and within-check statistics	Reading
`r6_boundary_sensitivity_v1.csv`	Focal threshold contrast: baseline cutpoints 0.54 and 0.43 imply 36.8% Injury, 50.0% Rupture, and 13.2% Distrust; lowering the upper cutpoint to 0.53 moves Rupture to 43.0% and Distrust to 20.2%	The baseline boundary is locally stable, but the fine Rupture-versus-Distrust split is threshold-sensitive

Table 13: R7. Alternative aggregation.

Full result table	Key rows and within-check statistics	Reading
`r7_alternative_aggregation_v1.csv`	Same-sign rate 54.0%; median absolute drift 0.01106; focal contrast: authority-stance severity stays positive at 0.0469, 0.0680, and 0.0494 across weights, while how severity turns negative under rake and unweighted aggregation	Alternative aggregation is a severe estimand-and-sample change rather than a routine replacement for the preferred narrator-level design

Table 14: R8. Minority-signal outcomes.

Full result table	Key rows and within-check statistics	Reading
`r8_minority_signal_v1.csv`	Same-sign rate 100.0%; median absolute drift 0; focal row: joint distrust-discomposure severity remains 0.1393 under IPW, 0.1287 under rake, and 0.0788 unweighted, with zero drift relative to the reference rows	The headline severity pattern does not depend on dropping minority-signal outcome cells

Table 15: R9. ILR dimensionality.

Full result table	Key rows and within-check statistics	Reading
`r9_ilr_dimensionality_v1.csv`	7 raw $p < 0.05$ rows; focal rows fall on latent basis coordinates rather than main-text outcomes, for example rake `ilr_joint6_5` severity = -3.611 ($p = 0.013$) and IPW `ilr_joint6_4` severity = 2.843 ($p = 0.043$)	Changing ILR dimensionality redistributes signal across latent coordinates, but does not generate a contradiction on the main authority/distrust margins

Table 16: R10. Omitted-confounding sensitivity.

Full result table	Key rows and within-check statistics	Reading
`r10_unobserved_confounding_sensitivity_v1.csv`	Median $RV_{q=1}$ across the panel is 3.2%; focal rows: IPW authority-stance severity has $RV_{q=1} = 0.1199$ and $RV_{q=1,\alpha=0.05} = 0.0482$, while the joint distrust-discomposure severity row has 0.1176 and 0.0458	The strongest severity rows are non-trivial, but they are not immune to omitted confounding

Table 17: R11. Restricted severity bundle.

Full result table	Key rows and within-check statistics	Reading
`r11_exogenous_severity_only_v1.csv`	Focal rows: total days incarcerated on authority stance = 0.000014 ($p = 0.0075$), 0.000014 ($p = 0.0073$), and 0.000010 ($p = 0.0082$); family separation on joint distrust-discomposure = 0.0570 ($p = 0.133$), 0.0655 ($p = 0.0589$), and 0.0662 ($p = 0.0459$)	The distrust-side result does not disappear when severity is decomposed, but no single component fully substitutes for the composite severity index

Table 18: R12. Interview-level long format.

Full result table	Key rows and within-check statistics	Reading
`r12_interview_long_format_v1.csv`	Same-sign rate 52.4%; sample is only 37 interviews representing 31 narrators; focal contrast: IPW authority-stance severity = 0.0724 ($p = 0.015$), while rake how severity = -0.2120 ($p = 0.010$)	The interview-level reformulation is too small and too unstable to replace the narrator-level analysis

Table 19: R13. Sequential DML.

Full result table	Key rows and within-check statistics	Reading
`r13_sequential_dml_v1.csv`	Same-sign rate 96.3%; median absolute drift 0.0007495; focal row: authority-stance timing = 0.0052 under sequential DML versus 0.0064 in the baseline unweighted fit	Sequential orthogonalization leaves the timing surface essentially unchanged

Table 20: R14. Placebo permutations.

Full result table	Key rows and within-check statistics	Reading
`r14_placebo_permutation_summary_v1.csv`	21 empirical $p < 0.05$ rows; focal contrast: authority-stance placebo p-values are 0, 0, and 0 across weights for age at first exposure, whereas how severity placebo p-values are 0.3125, 0.375, and 0.625	The placebo exercise is supportive for the strongest authority-side rows, but mixed elsewhere

Table 21: R15. Leave-one-camp-out.

Full result table	Key rows and within-check statistics	Reading
`r15_leave_one_camp_out_v1.csv`	Same-sign rate 74.9%; median absolute drift 0.006656; focal row: IPW authority-stance severity = 0.0362 without Minidoka, 0.0363 without Manzanar, and 0.0628 without Tule Lake	Facility leverage changes magnitudes, but the main severity result is not a single-site artifact

Table 22: R16. Recovered complete-case rerun.

Full result table	Key rows and within-check statistics	Reading
`r16_complete_case_v1.csv`	$N = 564$; focal rows: authority-stance severity = 0.0492, 0.0484, and 0.0424 across IPW, rake, and unweighted estimation, all with $p < 0.001$; joint distrust-discomposure severity = 0.1026 ($p = 0.0167$), 0.1059 ($p = 0.0114$), and 0.0895 ($p = 0.0089$)	The main severity result is not driven by recovered-missingness handling

Table 23: R17. Embedding arm versus baseline language-model arm.

Full result table	Key rows and within-check statistics	Reading
`r17_embedding_arm_v1.csv`	Same-sign rate 46.7%; median absolute drift 0.02048; focal rows sit on embedding-specific outcomes, including IPW `embed_share_Injury` severity = 0.0268 ($p = 0.0012$) and rake `embed_ilr_what_1` severity = 0.0831 ($p = 0.0034$)	The embedding arm carries its own signal, but aligns too weakly with the baseline arm to adjudicate the main results

Table 24: R18. Interview-year interaction.

Full result table	Key rows and within-check statistics	Reading
`r18_interview_year_interaction_v1.csv`	74 raw $p < 0.05$ rows overall; focal interaction rows on main-text outcomes are weak, with IPW authority stance = 0.0231 ($p = 0.790$), unweighted how = 0.1074 ($p = 0.666$), and unweighted joint distrust-discomposure = 0.1240 ($p = 0.188$)	Interview-year moderation does not rescue the weak timing result

Table 25: R19. Composure-separate model.

Full result table	Key rows and within-check statistics	Reading
`r19_composure_separate_v1.csv`	Same-sign rate 100.0%; median absolute drift 0; focal result: the displayed how coefficients have zero delta relative to the reference rows across IPW, rake, and unweighted estimation	The composure-only re-estimation leaves the how results unchanged

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